Chapter 4 Workshop

Moving Boundary Work

1. A frictionless piston–cylinder device contains 16 lbm of superheated water vapor at 40 psia and 600°F. Steam is now cooled at constant pressure until 70 percent of it, by mass, condenses. Determine the work done during this process, and draw the P-v diagram with respect to saturation lines.

Closed-System Energy Analysis: Ideal Gases

2. Nitrogen at 100 psia and 300°F in a rigid container is cooled until its pressure is 50 psia. Determine the work done and the heat transferred during this process, in Btu/lbm. ($C_v = 0.177 \frac{Btu}{lbm.R}$)

Closed System Energy Analysis

3. An insulated tank is divided into two parts by a partition. One part of the tank contains 2.5 kg of compressed liquid water at 60℃ and 600 kPa while the other part is evacuated. The partition is now removed, and the water expands to fill the entire tank. Determine the final temperature of the water and the volume of the tank for a final pressure of 10 kPa.